Abstract Algebra I (MAT5CJ302): syllabus weightage vs actual weightage

Paper: Fifth Semester CUFYUGP, November 2026 (2024 admission onwards), 70 marks, 2 hours. Syllabus: CU-FYUGP MAT5CJ302, text Fraleigh 7th ed. Questions are placed in modules by the syllabus item their content belongs to (source traced in `2026-NOV-paper-trace.pdf`).

Summary

- The paper follows the syllabus closely: no module is more than 3.6 percentage points from its share of teaching hours.
- **Module I was over-weighted** (28.6% of the marks on offer, against 25.0% of the hours). **Module IV was under-weighted** (21.4% against 25.0%).
- **Section C (10 marks) is all Module I and Module II.** Modules III and IV only appear in Sections A and B.
- Every module has at least 18 marks on offer in Sections A and B, above the syllabus minimum of 15 per module. Scaled to 70, Modules III and IV get exactly the minimum of 15.
- Module V (open-ended) carries no marks in the syllabus, and no question came from it. Q19b is printed in the 8th ed. inside §12 (Factor Groups), but its content (an isomorphism of G with itself) is Module I.

1. Syllabus weightage

Module	Syllabus items (7th ed.)	Hours	Share of hours (I–IV)	Marks rule (Ext 70)
I	§2 Binary operations, §3 Isomorphic binary structures, §4 Groups (items 1–6)	12	25.0%	Min. 15
II	§5 Subgroups, §6 Cyclic groups, §8 Groups of permutations (items 7–12)	14	29.2%	Min. 15
III	§9 Orbits, cycles, alternating groups; §10 Cosets and Lagrange (items 13–16)	10	20.8%	Min. 15
IV	§13 Homomorphisms, §18 Rings and fields, §19 Integral domains (items 17–22)	12	25.0%	Min. 15
V	Open-ended: generating sets, factor groups, field of quotients	12	—	No marks
Total		**48 + 12**	**100%**	**70**

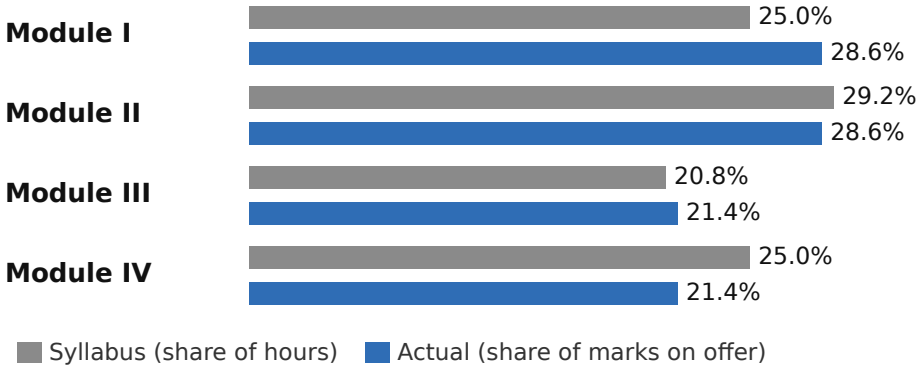
The syllabus gives no percentage weights, only hours and a minimum of 15 marks per module. "Syllabus weightage" below is each module's share of the 48 hours of Modules I–IV.

2. Actual weightage (November 2026)

Module	Section A (3 each)	Section B (6 each)	Section C (one of two)	Marks on offer	Share	Scaled to 70
I	Q1, Q2 = 6	Q11, Q12 = 12	Q19a, Q19b = 10	28	28.6%	20
II	Q3, Q4 = 6	Q13, Q14 = 12	Q20a, Q20b = 10	28	28.6%	20
III	Q5, Q6, Q7 = 9	Q15, Q16 = 12	—	21	21.4%	15
IV	Q8, Q9, Q10 = 9	Q17, Q18 = 12	—	21	21.4%	15
Total	**30 (ceiling 24)**	**48 (ceiling 36)**	**20 (answer 10)**	**98**	**100%**	**70**

"Marks on offer" counts every question printed, before the ceilings. Section C parts are counted as 5 + 5; the paper does not print the split. "Scaled to 70" = marks on offer × 70 / 98.

3. Syllabus vs actual



Module	Syllabus	Actual	Difference	Marks on offer, Sections A + B only	Meets min. 15?
I	25.0%	28.6%	**+3.6**	18	Yes
II	29.2%	28.6%	**−0.6**	18	Yes
III	20.8%	21.4%	**+0.6**	21	Yes
IV	25.0%	21.4%	**−3.6**	21	Yes

4. Question-by-question placement

Q	Marks	Topic	Syllabus item (7th ed.)	Module
1	3	Binary operation ab + 1: commutative, associative	§2 Binary operations (items 1–2)	I
2	3	Inverse of an isomorphism	§3 Isomorphic binary structures (items 3–4)	I
3	3	x + y = 0 as a subgroup of $\mathbb{R}^2$	§5 Subgroups (item 7)	II
4	3	Cyclic group, generators	§5–§6 Cyclic subgroups and groups (items 8–10)	II
5	3	Orbits of a permutation	§9 Orbits (item 13)	III
6	3	Cosets of {0,4} in $\mathbb{Z}_8$	§10 Cosets (item 15)	III
7	3	Subgroup orders for |G| = 24	§10 Lagrange (items 15–16)	III
8	3	Kernel of φ(n) = 6n	§13 Homomorphisms (items 17–18)	IV
9	3	Integral domain; $\mathbb{Z}_p$	§19 Integral domains (items 21–22)	IV
10	3	Finite integral domain is a field	§19 Integral domains (item 22)	IV
11	6	Group-table equations have unique solutions	§4 Group tables (item 6)	I
12	6	Even order ⇒ element of order 2	§4 Finite groups (item 6)	I
13	6	$ab^{-1} \in H$ is an equivalence relation	§5 Subgroups, exercise (item 7)*	II
14	6	Unique element of order 2	§6 Cyclic groups (items 9–10)	II
15	6	Square of an odd cycle	§9 Cycles (items 13–14)	III
16	6	No proper subgroups ⇒ prime order	§10 Lagrange (item 16)	III
17	6	Composite of homomorphisms	§13 Homomorphisms (item 17)	IV
18	6	Divisors of 0 in $\mathbb{Z}_n$	§19 Integral domains (item 21)	IV
19a	5	$(ab)^2 = a^2b^2 \Rightarrow$ abelian	§4 Groups (items 5–6)	I
19b	5	i_g(x) = gxg′ is an isomorphism	§3–§4 Isomorphism of groups (items 4–6)*	I

20a	5	S_A is a group	§8 Groups of permutations (items 11–12)*	II
20b	5	|ab| = n ⇒ |ba| = n	§6 Cyclic groups, order of an element (items 9–10)	II

* Placements that could go another way: Q13 is a §5 exercise but is the coset relation of §10 (Module III); Q19b's text in the 8th ed. is in §12, which is Module V by section but Module I by content; Q20a is 8th-ed Theorem 4.8, which the study folders file under Module III, but the 7th-ed syllabus places S_A in §8 (Module II). If Q13 and Q20a are moved to Module III, the shares become I 28.6%, II 17.3%, III 32.7%, IV 21.4%. The overall finding is unchanged: Module IV is the lightest, and Section C contains nothing from Module IV.